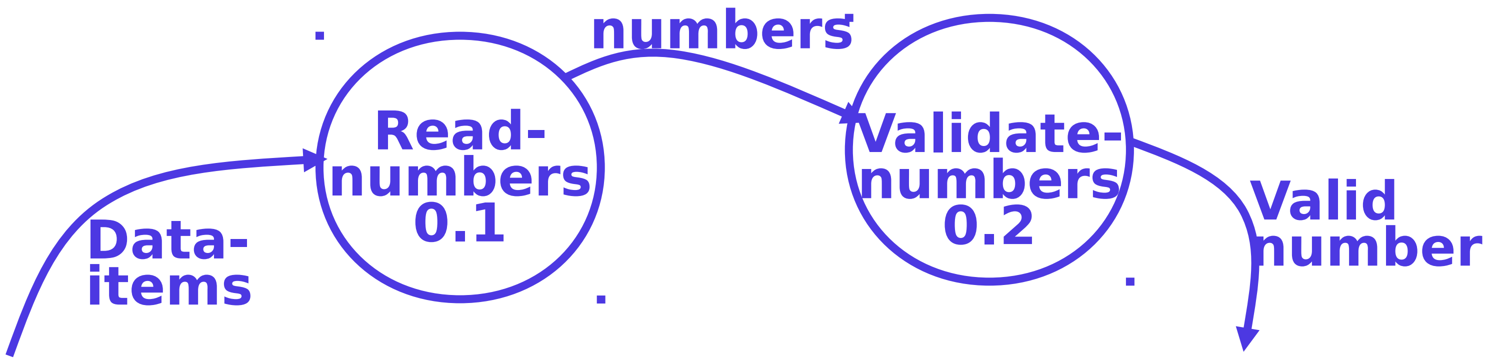
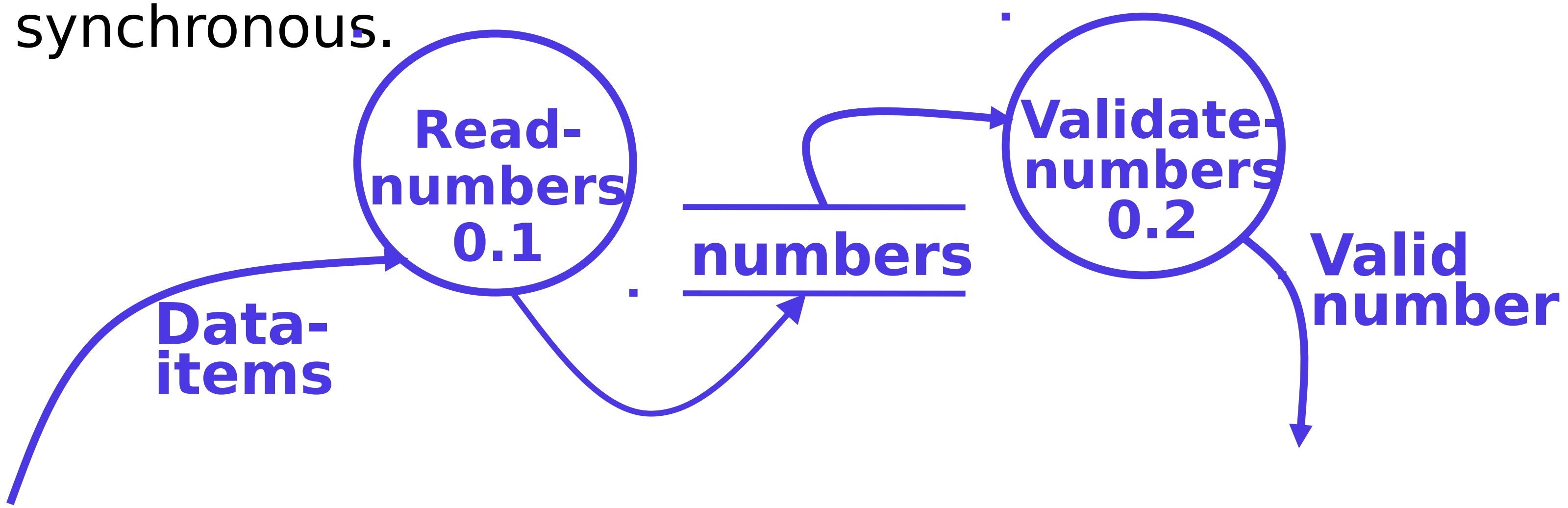
Function-Oriented Software Design

DFD Operations

- Synchronous Operation
  - If two bubbles are directly connected by a data flow arrow, they are said to be synchronous.



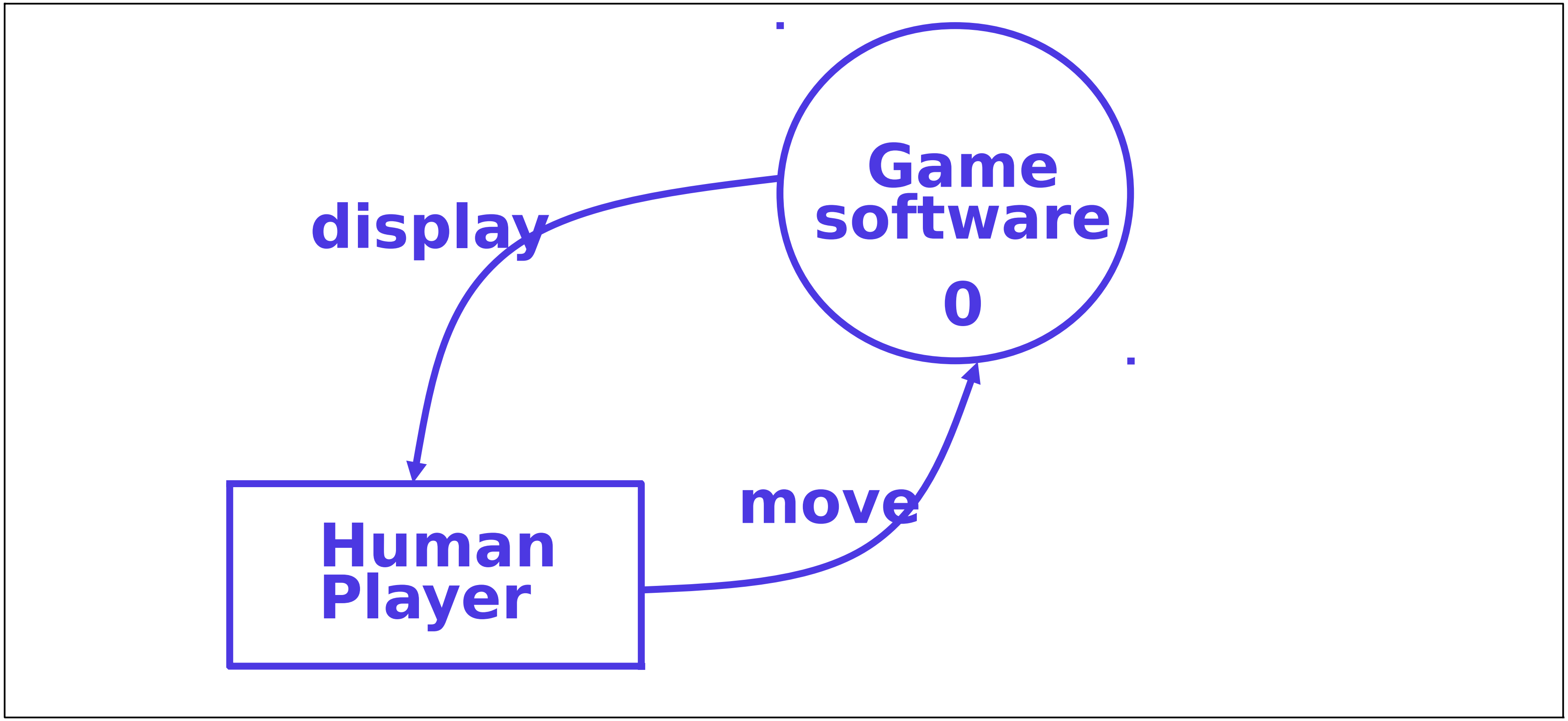
- Asyncrhonous Operation
  - If two bubbles are connected via a data store, they are not synchronous.



How is Structured Analysis Performed?

- Initially represent the software at the most abstract level:
  - Called the Context diagram.
  - The entire system is represented as a single bubble,
  - This bubble is labelled according to the main function of the system.
- A context diagram shows:
  - Data input to the system, Output data generated by the system, and External entities.
- Context diagram captures:
  - Various entities external to the system and interacting with it.
  - Data flow occurring between the system and the external entities.
- The context diagram is also called as the level 0 DFD.
- Establishes context of the system, & is used to represent
  - Data sources
  - Data sinks

Game Software: Context Diagram

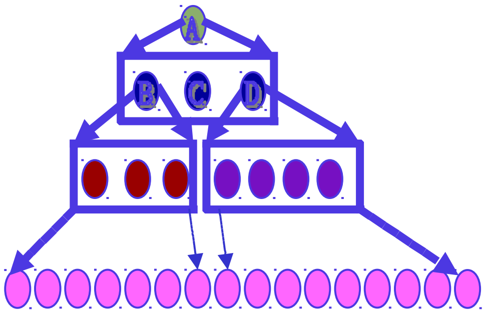


How to construct a Level 1 DFD ?

- Examine the SRS / Requirements document:
  - Represent each high-level function as a bubble.
  - Represent data input to every high-level function.
  - Represent data output from every high-level function.

Higher Level DFDs

- Each high-level function is separately decomposed into subfunctions:
  - Identify the subfunctions of the function
  - Identify the data input to each subfunction
  - Identify the data output from each subfunction



- These are represented as subsequent level DFDs.

Decomposition

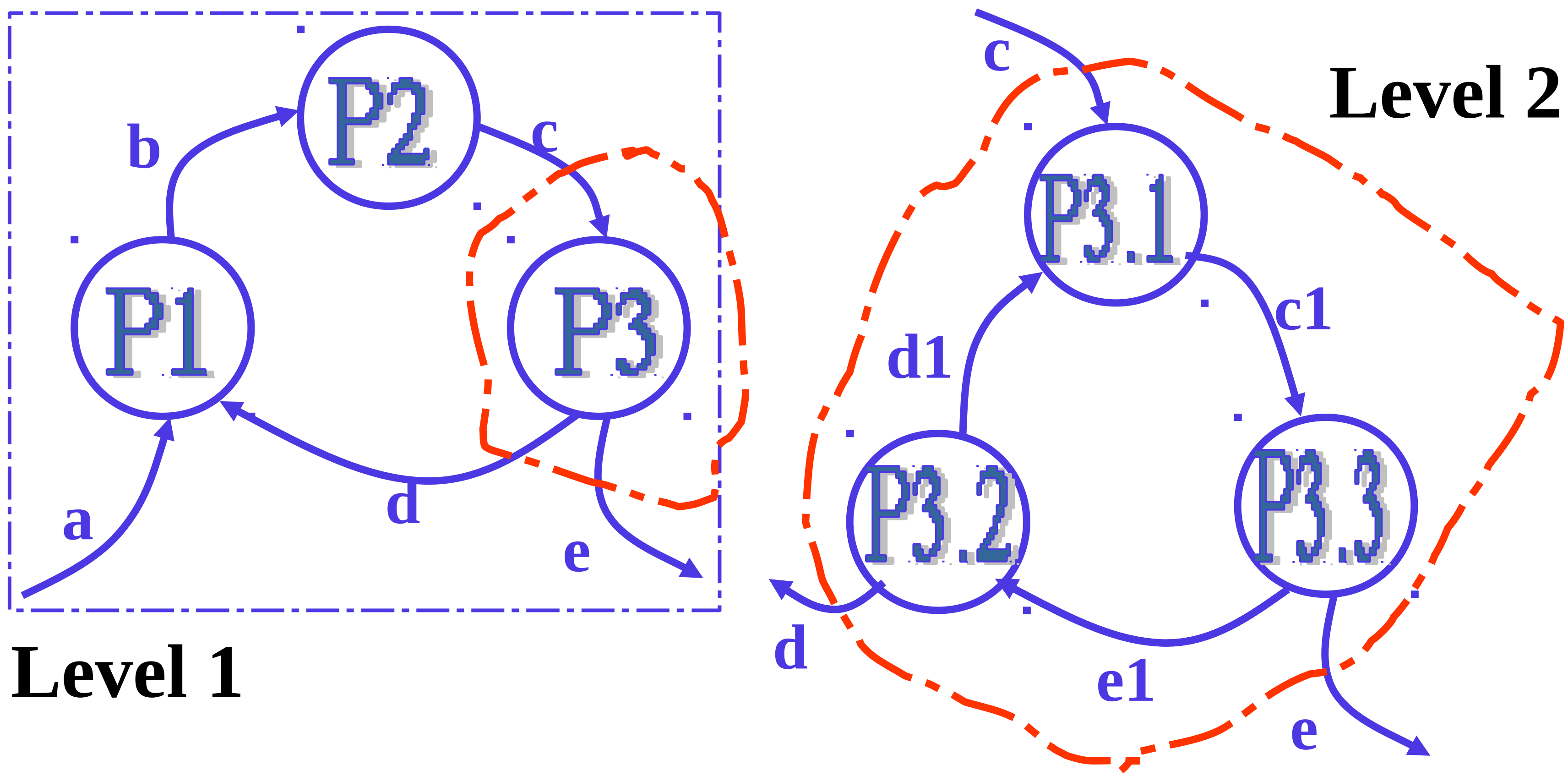
- Decomposition of a bubble - Also called factoring or exploding.
- Each bubble is decomposed to between 3 to 7 bubbles.
- Too few bubbles make decomposition superfluous:
  - If a bubble is decomposed to just one or two bubbles, then this decomposition is redundant.
- Too many bubbles:
  - More than 7 bubbles at any level of a DFD.
  - Make the DFD model hard to understand.
- Decompose How Long?
  - Decomposition of a bubble should be carried on until:
    - A level at which the function of the bubble can be described using a simple algorithm.

Numbering of Bubbles

- Number the bubbles in a DFD:
  - Numbers help in uniquely identifying any bubble from its bubble number.
- The bubble at context level:
  - Assigned number 0.
- Bubbles at level 1:
  - Numbered 0.1, 0.2, 0.3, etc
- When a bubble numbered x is decomposed,
  - Its children bubble are numbered x.1, x.2, x.3, etc.

Balancing a DFD

- Data flowing into or out of a bubble Must match the data flows at the next level of DFD.
- In the level 1 of the DFD, data item c flows into the bubble P3 and the data item d and e flow out.
- In the next level, bubble P3 is decomposed.
  - The decomposition is balanced as data item c flows into the level 2 diagram and d and e flow out.



Data Dictionary

- A DFD is always accompanied by a data dictionary.
- A data dictionary lists all data items appearing in a DFD:
  - Definition of all composite data items in terms of their component data items.
  - All data names along with the purpose of the data items.
- For example, a data dictionary entry may be:
  - grossPay = regularPay+overtimePay
- Significance of Data Dictionary
  - Provides a standard terminology for all data in a project.
  - A consistent vocabulary for data is very important
    - Different team members tend to use different terms to refer to the same data, and it causes unnecessary confusion.
- Data dictionary provides the definition of different data, in terms of their component elements.
  - For large systems,
    - The data dictionary grows rapidly in size and complexity.
    - Typical projects can have thousands of data dictionary entries.
    - It is extremely difficult to maintain such a dictionary manually.

Semantics for Data Definition

- Composite data are defined in terms of primitive data items using following operators:
  - +: denotes composition of data items, e.g
    - a+b represents data a and b.
  - [,,,]: represents selection, i.e. any one of the data items listed inside the square bracket can occur.
    - For example, [a,b] represents either a occurs or b occurs.
  - = represents equivalence,
    - e.g. a=b+c means that a represents b and c.
  - * *: Anything appearing within * * is a comment.
  - ( ): contents inside the bracket represent optional data
    - which may or may not appear.
    - a+(b) represents either a or a+b occurs.
  - {}: represents iterative data definition,
    - {name}5 represents five name data.
    - {name}* represents zero or more instances of name data.

Observation

- DFDs help us create Data model as well as Function model
- As a DFD is refined into greater levels of detail:
  - The analyst performs an implicit functional decomposition.
  - At the same time, refinements of data takes place.
- Common mistakes committed / Key guidelines,
  - Drawing more than one bubble in context diagram.
  - All external entities should be represented in context diagram.
  - External entities should not appear at any other level of DFD.
  - Only 3 to 7 bubbles per DFD level should be allowed.
  - Each bubble should be decomposed to between 3 and 7 bubbles.
  - Attempting to represent control information in a DFD.
    - trying to represent the order in which different functions are / will be executed.
  - All functions of the system must be captured in the DFD model - No function specified in the SRS document should be omitted.
  - Only those functions specified in SRS document should be represented.
    - Do not assume extra functionality of the system not specified by the SRS document.

Represent data that flows between bubbles and not the conditions depending on which a process is invoked.